Computational Logic

Deduction

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Previously on LC...

We have define:

- The language of propositional logic
- The semantics of propositional logic

What is “a logic”?

- a language of an alphabet in which it is defined what are **formulas** and other objects.
- A way to define semantics (=notion of what is true) for the formulas. The same logic can have several semantic interpretations...
- A deduction system that, using a finite number of rules allows us to obtain formulas from others (=reasoning). Must be sound: what if deduces must be true... But it will also be good that what is true must be deducible (complete)... Although this is not always possible.

Semantics versus deduction

In propositional logic:

Semantics		**Deduction**
	$\xrightarrow{(complete)}$	
$\Gamma \models \phi$		$\Gamma \vdash \phi$
	$\xleftarrow{(sound)}$	

Reasoning

1	If Paul falls asleep and someone wakes him up, he swears
2	Paul fell asleep
3	He didn't say bad words
4	Nobody woke him

Is it a valid reasoning?

Reasoning

General form of the above reasoning:

1	$(p \land q) \to r$
2	p
3	$\neg r$
4	$\neg q$

We are only interested in the logical structure of sentences.

Valid and sound reasoning

Informally a reasoning is a sequence of statements of which one of them – the conclusion – must be consequence of the rest – the premises.

The conclusion is a logical consequence of the premises, if it is true whenever the premises are true...

In this case we have a valid reasoning.

In a valid reasoning, if the premises are true, the reasoning is sound.

The reasoning we saw was valid and sound...

Another reasoning

1	If you fail, you have to repeat
2	If you don't study, you fail
3	You are not repeating
4	Study or Fail or Both

It is also valid.

Formally

1	$r \to t$
2	$\neg e \to r$
3	$\neg t$
4	$e \vee r \vee (e \wedge r)$

Deduction methods

But how do we show that a conclusion is a logical consequence of the premises?

Building a succession of steps in which, at each one, the conclusion is a consequence of previous conclusions and premises.

Formally we will define deduction systems

On the other hand, to show that a conclusion is not a logical consequence of the premises, we have to show that there is a situation where the premises can be true and the conclusion false. This situation is called a counterexample.

Deduction versus Semantics

It is *"easy"* to see that if a formula ϕ is a logical consequence of a set of formulas Γ then $\Gamma \models \phi$ i.e. ϕ is a semantic consequence of Γ.

Why use deduction systems?

To have an automatic or at least semi-automatic method to obtain the validity of a formula or its consequence from others.

A deduction is a syntactic method.

Axiomatic deduction systems

In the beginning of the 20th century, Hilbert introduced the basis for modern proof theory.



Axiomatic deduction system $\mathcal{D}$ - It consists of:

- axioms (logical) base formulas that translate the connective properties
- inference rules way to get formulas from others

Axiomatic deduction systems - deduction

Deduction, $\Sigma \vdash_{\mathcal{D}} \phi_n$ - A finite succession of formulas $\phi_1, \ldots \phi_n$ is a deduction of ϕ_n in $\mathcal{D}$ from a set of formulas Σ if for each $1 \leq i \leq n$ it is verified:

- $\phi_i \in \Sigma$, or
- ϕ_i is an axiom, or
- ϕ_i results from $\phi_1 \ldots \phi_{i-1}$ by applying an inference rule

And write $\Sigma \vdash_{\mathcal{D}} \phi_n$

Deduction Systems

Theorem and Proof: If $\Sigma = \emptyset$, then ϕ_n is a theorem (from $\mathcal{D}$) is written $\vdash_{\mathcal{D}} \phi_n$.

In this case, the deduction $\phi_1, \ldots \phi_n$ is said to be a proof of ϕ_n.

If $\Sigma = \{\theta_1, \ldots, \theta_n\}$ is finite, instead of $\Sigma \vdash_{\mathcal{D}} \phi$ we write

$$\theta_1, \ldots, \theta_n \vdash_{\mathcal{D}} \phi$$

and $\mathcal{D}$ will be omitted if it is clear from the context.

$$\theta_1, \ldots, \theta_n \vdash \phi$$

Natural Deduction System, *DN*

System invented by G. Gentzen (1935) and whose rules are intended to reflect the forms of reasoning used in the mathematical proofs.



- allows for the introduction of hypotheses in the middle of the deduction
- has no axioms... only inference rules

For each logical connective there are two types of rules: introduction and elimination. We start with formulas with the following connectives: $\wedge$, $\vee$, $\neg$, $\mathbf{F}$ and $\rightarrow$

Example

How to show that a formula is valid (i.e. a tautology)?

$$(p \lor (q \land r)) \to ((p \lor q) \land (p \lor r))$$

$$\begin{array}{c|c} p & q \land r \\ p \lor q & q \\ p \lor r & r \\ ((p \lor q) \land (p \lor r)) & p \lor q \\ & p \lor r \\ & ((p \lor q) \land (p \lor r)) \end{array}$$

Rule representation

A inference rule is of the form:

$$\text{from } \phi_1 \dots \phi_k \text{ infer } \phi$$

And it can be graphically represented by:

$$\frac{\phi_1, \dots, \phi_k}{\phi}$$

As a deduction is a sequence of applications of inference rules it can be represented by a tree.

DN Inference Rules: Conjunction

Introduction

$$\frac{\phi \qquad \psi}{\phi \ \wedge \ \psi} \wedge \mathbf{I}$$

If we already deduced ϕ and ψ then we can deduce $\phi \ \wedge \ \psi$

Elimination

$$\frac{\phi \ \wedge \ \psi}{\phi} \wedge \mathbf{E_1} \qquad \frac{\phi \ \wedge \ \psi}{\psi} \wedge \mathbf{E_2}$$

If we deduce $\phi \ \wedge \ \psi$ we can deduce ϕ; and we can also deduce ψ.

Example

Show that

$$p \wedge q, r \vdash q \wedge r$$

We can build the deduction in a *tree*:

$$\dfrac{\dfrac{p \wedge q}{q} \wedge \mathbf{E}_2 \qquad r}{q \wedge r} \wedge \mathbf{I}$$

Where the leaves are the premises...

But these trees can get very big...

Fitch Notation



Let us consider a linear representation for deductions: we number the steps and indicate which rule to apply and which formulas are involved:

1	$p \wedge q$	
2	r	
3	q	$\wedge$E, 1
4	$q \wedge r$	$\wedge$I, 3, 2

Fitch Notation $\wedge$

Introduction

$$\begin{array}{|l}
\phi \\
\vdots \\
\psi \\
\vdots \\
\phi \ \wedge \ \psi \qquad \wedge\mathsf{I}
\end{array}$$

Elimination

$$\begin{array}{|l}
\vdots \\
\phi \ \wedge \ \psi \\
\vdots \\
\phi \qquad \wedge\mathsf{E}
\end{array}$$

$$\begin{array}{|l}
\vdots \\
\phi \ \wedge \ \psi \\
\vdots \\
\psi \qquad \wedge\mathsf{E}
\end{array}$$

Example

Show that

$$(p \land q) \land r \vdash r \land q$$

1	$(p \land q) \land r$	
2	$p \land q$	$\land$E, 1
3	r	$\land$E, 1
4	q	$\land$E, 2
5	$r \land q$	$\land$I, 3, 4

Rules of inference *DN*: Disjunction

Introduction

$$\frac{\phi}{\phi \ \vee \ \psi} \vee \mathbf{I_1} \qquad \frac{\psi}{\phi \ \vee \ \psi} \vee \mathbf{I_2}$$

If we already deduced ϕ we can deduce any disjunction having ϕ.

Elimination

$$\frac{\phi \ \vee \ \psi \qquad \begin{matrix}[\phi] \\ \vdots \\ \gamma\end{matrix} \qquad \begin{matrix}[\psi] \\ \vdots \\ \gamma\end{matrix}}{\gamma} \vee \mathbf{E}$$

- if we have already deduced the disjunction $\phi \ \vee \ \psi$
- if assuming ϕ we deduce γ (in a sub-deduction)
- and if assuming ψ we deduce γ (in a sub-deduction)
- then we can deduce γ

Fitch Notation $\vee$

Introducion

$$\begin{array}{l} \vdots \\ \phi \\ \vdots \\ \phi \vee \psi \end{array} \vee\text{I} \qquad \begin{array}{l} \vdots \\ \phi \\ \vdots \\ \psi \vee \phi \end{array} \vee\text{I}$$

Elimination

$$\begin{array}{l} \phi \vee \psi \\ \vdots \\ \quad \begin{array}{|l} \phi \\ \hline \quad \vdots \\ \quad \gamma \end{array} \\ \quad \begin{array}{|l} \psi \\ \hline \quad \vdots \\ \quad \gamma \end{array} \\ \gamma \end{array} \vee\text{E}$$

Example

Show that $(p \land q) \lor (q \land r) \vdash q$.

1	$(p \land q) \lor (q \land r)$	
2	$\quad p \land q$	
3	$\quad q$	$\land$E, 2
4	$\quad q \land r$	
5	$\quad q$	$\land$E, 4
6	q	$\lor$E, 1, 2–3, 4–5

Repetition

In a deduction we can always repeat a conclusion already reached.

We will call this rule repetition:

$$\frac{\phi}{\phi}\mathbf{R}$$

Show that $(p \wedge q) \vee q \vdash q$

1	$(p \wedge q) \vee q$	
2	$\quad p \wedge q$	
3	$\quad q$	$\wedge$E, 2
4	$\quad q$	
5	$\quad q$	R, 4
6	q	$\vee$E, 1, 2–3, 4–5

Using sub-deductions

1	$(p \wedge q) \vee (q \wedge r)$	
2	$\quad p \wedge q$	
3	$\quad p$	$\wedge$E, 2
4	$\quad q$	$\wedge$E, 2
5	$\quad q \wedge r$	
6	$\quad q$	$\wedge$E, 5
7	q	$\vee$E, 1, 2–4, 5–6
8	$q \wedge p$	$\wedge$I, 7, 3

This deduction is WRONG! In step 8 a formula that was deducted in a sub-deduction that has already ended.

VERY IMPORTANT!!!

One sub-deduction starts with the introduction of new hypotheses (premises) and the deductions made there depend on them. When the sub-deduction ends, these hypotheses are no longer assumed and therefore these hypothesis (and their conclusions) cannot be used.

DN Inference Rules: Negation

Elimination It corresponds to one of the parts of the double negation principle.

$$\frac{\neg\neg\phi}{\phi}\neg\mathbf{E}$$

Introduction

This rule corresponds to proofs by contradiction. We represent by $\mathbf{F}$ a contradiction (eg $\phi \wedge \neg\phi$).

$$\begin{array}{c} [\phi] \\ \vdots \\ \mathbf{F} \end{array}\over{\neg\phi}\neg\mathbf{I}$$

If assuming ϕ we can deduce a contradiction, then we can deduce $\neg\phi$ from the original assumptions.

Inference Rules *DN*: F

If we do not consider **F** as an abbreviation of $\phi \wedge \neg\phi$, we must have a rule to introduce it:

Introduction $(*)$

$$\dfrac{\begin{array}{c}\phi \\ \vdots \\ \neg\phi\end{array}}{\mathbf{F}}\ \mathbf{F\ I}$$

If we deduce ϕ and $\neg\phi$ then we have a contradiction.

Example

Show $\phi \vdash \neg\neg\phi$

1	ϕ	
2	$\quad\neg\phi$	
3	$\quad\mathbf{F}$	$\mathbf{F}\ \mathbf{I}$, 1, 2
4	$\neg\neg\phi$	$\neg$I, 2–3

Rules of inference *DN*: F

Elimination

$$\frac{\mathbf{F}}{\phi}\,\mathbf{F}\ \mathbf{E}$$

If we deduce a contradiction we can deduce any formula.

A set of formulas Σ is said to be inconsistent if $\Sigma \vdash \mathbf{F}$.

Fitch notation ¬ and F

Introduction	Elimination
¬	ϕ / ⋮ / **F** / $\neg\phi$ ¬I ⋮ / $\neg\neg\phi$ / ⋮ / ϕ ¬E
F	ϕ / ⋮ / $\neg\phi$ / ⋮ / **F** **F I** ⋮ / **F** / ⋮ / ϕ **F E**

Inference Rules *DN*: Implication

Elimination

This rule is commonly known as modus ponens (from the Latin) and corresponds to conditional reasoning:

$$\frac{\phi \quad \phi \to \psi}{\psi} \to \mathbf{E}$$

If we already deduced ϕ and $\phi \to \psi$ then we can deduce ψ.

Example

Show that $p, p \to q, p \to (q \to r) \vdash r$:

1	$p \to (q \to r)$	
2	$p \to q$	
3	p	
4	$q \to r$	$\to$E, 1, 3
5	q	$\to$E, 2, 3
6	r	$\to$E, 4, 5

Inference Rules *DN*: Implication

Introduction (rule of deduction)

The rule for introducing an implication needs a sub-deduction: assuming ϕ we try to deduct ψ. If this happens, we finish the sub-deduction (taking the assumption) and conclude $\phi \to \psi$:

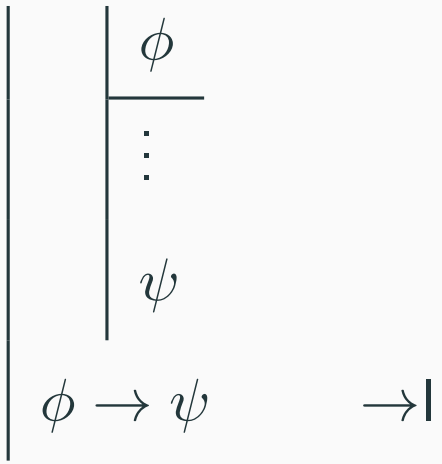
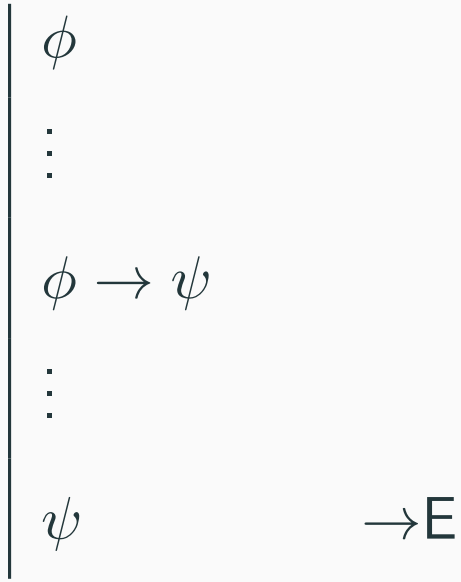
$$\dfrac{\begin{array}{c}[\phi] \\ \vdots \\ \psi\end{array}}{\phi \to \psi} \to \mathbf{I}$$

Example

Show that $(p \lor q) \to r \vdash p \to r$:

1	$(p \lor q) \to r$	
2	$\quad p$	
3	$\quad p \lor q$	$\lor$I, 2
4	$\quad r$	$\to$E, 1, 3
5	$p \to r$	$\to$I, 1, 2–4

Fitch notation $\rightarrow$

Introduction	Elimination
	

Deductions without premises

With the introduction of implication (rule of deduction) we can convert any deduction with premises into a deduction without premises:

Show $\vdash \phi \to \neg\neg\phi$

1	ϕ	
2	$\neg\phi$	
3	**F**	**F I**, 1, 2
4	$\neg\neg\phi$	$\neg$I, 2–3
5	$\phi \to \neg\neg\phi$	$\to$I, 1–4

Examples

Show using DN:

(a) $p \wedge (q \vee r) \vdash (p \wedge q) \vee (p \wedge r)$

(b) $\neg p \vee \neg q \vdash \neg(p \wedge q)$

(c) $\vdash \neg(p \wedge \neg p)$

(d) $\neg(\neg p \vee q) \vdash p$